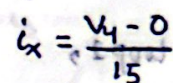


Section: ~~01A~~ 1A

sat sun mon tue wed thu fri Date:



$$i_x = -3 \text{ mA}$$

$$A \cap N = \{ \} \quad \Leftarrow$$

$V_{12}(KCL)$

$$V_1\left(\frac{1}{4}\right) - \frac{V_3}{4} - 5i_x = 0$$

$$\Rightarrow V_1 \cdot \frac{1}{4} - \frac{V_3}{4} - 5 \cdot \frac{V_4}{15} = 0$$

$$\Rightarrow \frac{V_1}{4} - \frac{V_3}{4} - \frac{V_4}{3} = 0$$

$$\Rightarrow 3V_1 - 3V_3 - 4V_4 = 0 \Rightarrow$$

$$\Rightarrow 3v_1 - 3v_4 - 30 - 4v_4 = 0 \Rightarrow 3v_1 - 7v_4 = 30$$

KCL (V₅)

$$V_5 \left(\frac{1}{2} + \frac{1}{6} + \frac{1}{5} \right) - \frac{V_2}{2} - \frac{V_4}{6} - \frac{0}{5} = 0$$

$$\Rightarrow \frac{4}{5}V_5 - \frac{V_2}{2} - \frac{V_3}{5} = 0$$

$$\Rightarrow -5V_2 - V_4 + 8V_5 = 0 \quad \text{--- (11)}$$

Solving,

$$V_1 = -95 \text{ V} \quad V_2 = 65 \text{ V} \quad V_4 = -45 \text{ V}$$

$$V_5 = 35 \text{ V} \quad V_3 = -35 \text{ V}$$

$$V_6 = 5V \quad V_7 = 0V$$

$$0 = (KCL \ V_2) + (P - \epsilon_j) 2(1 + \epsilon_j) 2 =$$

$$V_2\left(\frac{1}{3}\right) - \frac{V_5}{2} + 5 \cdot \frac{V_4}{15} = 0$$

$$\Rightarrow \frac{V_2}{2} - \frac{V_5}{2} + \frac{V_4}{3} = 0$$

$$\Rightarrow 3V_2 + 2V_4 - 3V_5 = 0 \quad \text{--- (11)}$$

KCL (V_3 & V_4 superNode),

$$v_3(\frac{1}{4} + \frac{1}{10}) - \frac{v_1}{4} - \frac{5}{10} + v_4(\frac{1}{10} + \frac{1}{15}) - \frac{v_5}{10} - \frac{0}{15} = 0$$

$$\Rightarrow \frac{7}{20} v_3 - \frac{v_1}{4} - \frac{1}{2} + \frac{v_4}{6} - \frac{v_5}{10} = 0$$

$$\Rightarrow -5V_1 + 7V_3 + \frac{10}{3}V_4 - 2V_5 = 10$$

$$\Rightarrow -15V_1 + 21V_3 + 10V_4 - 6V_5 = 30$$

KVL (supernode)

$$V_3 - V_4 = W$$

$$\Rightarrow V_3 = V_4 + 10$$

$$\Rightarrow -15V_1 + 21V_1 + 210 + 10V_1 - 6V_5 = 30$$

$$\Rightarrow -15V_1 + 31V_4 - 6V_5 = -180$$

Mesh Analysis.

loop 1,

$$i_1 = -5i_2$$

$$\Rightarrow i_1 + 5i_2 - 5i_3 = 0 \quad \text{--- (i)}$$

loop 2 (KVL),

$$-5 + 10i_2 + 10 + 15i_2 - 15i_3 = 0$$

$$\Rightarrow 25i_2 - 15i_3 = -5 \quad \text{--- (ii)}$$

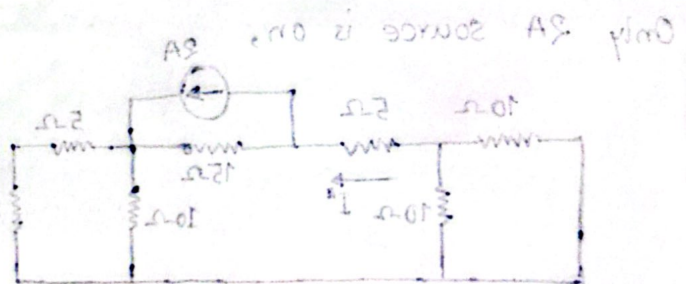
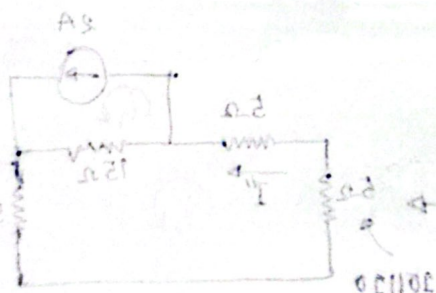
loop 3 (KVL),

$$5i_3 + 15i_3 - 15i_2 + 10i_3 - 10i_1 = 0$$

$$\Rightarrow -10i_1 - 15i_2 + 30i_3 = 0 \quad \text{--- (iii)}$$

Solving (i), (ii) & (iii)

$$i_1 = 15\text{mA}; \quad i_2 = 4\text{mA}; \quad i_3 = 7\text{mA}$$



$$I_1 + I_2 = I \quad \text{--- (iv)}$$

$$I_1 + I_2 =$$

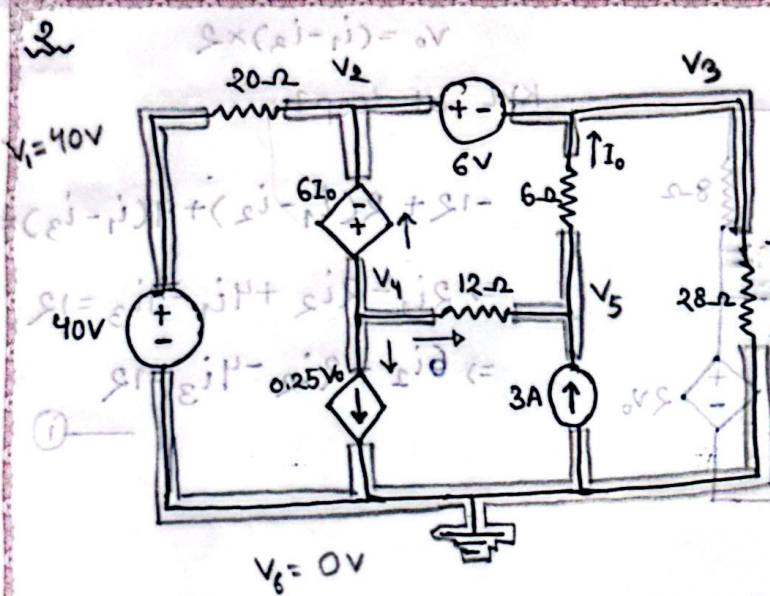
$$I_1 + I_2 =$$

$$(10\text{mA})$$

$$0 = i_1^2 + (2-i_1)^2 + i_1^2 + i_1^2 + 10(i_1^2 - 2) + 2i_1^2 = 0$$

$$AI = i_1^2 \Rightarrow 30i_1^2 - 30 = 0 \Rightarrow i_1 = 1\text{A}$$

$$AI = i_1^2 = 1$$



$$I_0 = \frac{V_5 - V_3}{6}$$

$$V_6 = V_3 - 0 \Rightarrow V_6 = V_3$$

$$V_5 \left(\frac{1}{6} + \frac{1}{12} \right) - \frac{V_3}{6} - \frac{V_4}{12} - 3 = 0$$

$$\Rightarrow \frac{V_5}{4} - \frac{V_3}{6} - \frac{V_4}{12} = 3$$

$$\Rightarrow -2V_3 - V_4 + 3V_5 = 36 \quad \text{--- (1)}$$

Node 2, 3 & 4 are supersuper Node

KCL,

$$V_2 \left(\frac{1}{20} \right) - \frac{40}{20} + V_3 \left(\frac{1}{28} + \frac{1}{6} \right) - \frac{0}{28} - \frac{0}{6} + V_4 \left(\frac{1}{12} \right) - \frac{12}{12} \frac{V_5}{12} + 0.25V_6 = 0$$

$$\Rightarrow \frac{V_2}{20} - 2 + \frac{17}{84} V_3 - \frac{V_5}{4} + \frac{V_4}{12} + \frac{V_3}{14} = 0$$

$$\Rightarrow \frac{V_2}{20} + \frac{23}{84} V_3 + \frac{V_4}{12} - \frac{V_5}{4} = 2 \quad \text{--- (II)}$$

KVL,

$$V_2 - V_3 = 6 \quad \text{--- (III)}$$

$$V_4 - V_2 = 6I_0$$

$$\Rightarrow -V_2 + V_4 = V_5 - V_3$$

$$\Rightarrow -V_2 + V_3 + V_4 - V_5 = 0 \quad \text{--- (IV)}$$

$$I = I + 0.25V_6 + \frac{V_4 - V_5}{12} = 0$$

$$\Rightarrow I = -\frac{56.91 - 50.91}{12} - 0.25 \times 29.91$$

$$\therefore I = -7.9775 \text{ A}$$

Solving (I), (II), (III) and (IV)

$$V_2 = 35.91 \text{ V} \quad V_3 = 29.91 \text{ V}$$

$$V_4 = 56.91 \text{ V} \quad V_5 = 50.91 \text{ V}$$

$$V_1 = 40 \text{ V} \quad V_6 = 0 \text{ V}$$

$$P = VI$$

$$= 6I_0 \times \frac{V_5 - V_3}{6}$$

$$P = VI$$

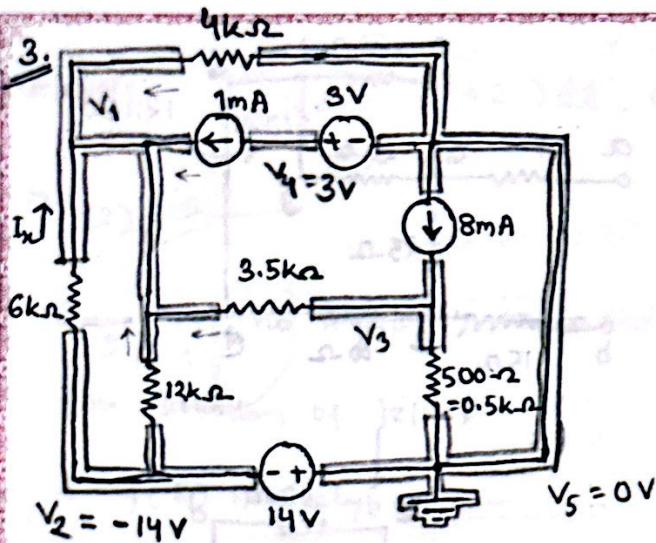
$$= 6I_0 \times 1 \text{ V}$$

$$= 6 \times \frac{V_5 - V_3}{6} \times 1 = (50.91 - 29.91) \times$$

$$= -167.5275 \text{ W}$$

(Supplying)

$$(-7.9775)$$



(a) Node Analysis would be easier. There are supermesh but no super Node and we can easily compute 3 Node without any equation. We need only two equations for V_1 and V_3 .

For V_1 , (KCL)

$$V_1 \left(\frac{1}{4} + \frac{1}{3.5} + \frac{1}{12} + \frac{1}{6} \right) - \frac{0}{4} - \frac{V_3}{3.5} - \frac{-14}{12} - \frac{-14}{6} - 1 = 0$$

$$\Rightarrow \frac{11}{14} V_1 - \frac{V_3}{3.5} + \frac{5}{2} = 0$$

$$\Rightarrow 11V_1 - 4V_3 = -35 \quad \text{--- (1)}$$

For V_3 , (KCL)

$$V_3 \left(\frac{1}{3.5} + \frac{1}{0.5} \right) - \frac{V_1}{3.5} - \frac{0}{0.5} - 8 = 0$$

$$\Rightarrow \frac{16}{7} V_3 - \frac{V_1}{3.5} = 8$$

$$\Rightarrow -2V_1 + 16V_3 = 56 \quad \text{--- (2)}$$

Solving (1) and (2).

$$V_1 = -2V \quad V_3 = 3.25V$$

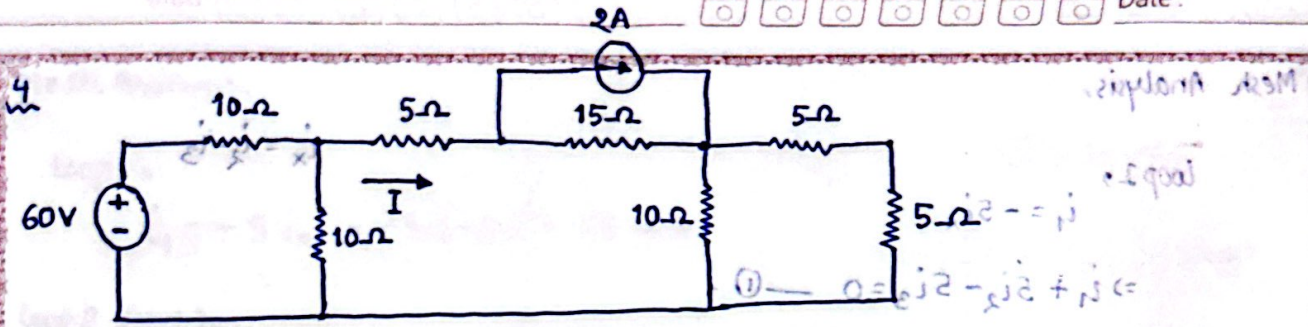
$$V_2 = -14V \quad V_4 = 3V, \quad V_5 = 0V$$

(b)

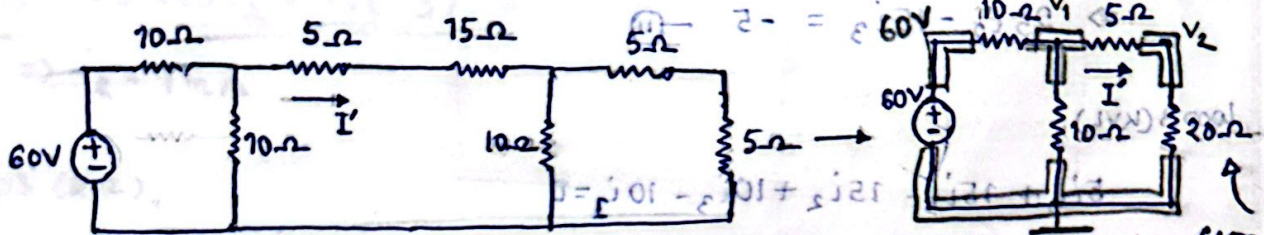
$$I_x + \frac{V_5 - V_1}{4} + 1 + \frac{V_3 - V_1}{3.5} + \frac{V_2 - V_1}{12} = 0$$

$$\Rightarrow I_x + \frac{0 + 2}{4} + 1 + \frac{3.25 + 2}{3.5} + \frac{-14 + 2}{12} = 0$$

$$\Rightarrow I_x = -2A = -2mA$$



Only 60V source is on,



KCL on V_1

$$V_1 \left(\frac{1}{10} + \frac{1}{10} + \frac{1}{5} \right) - \frac{60}{10} - \frac{V_2}{5} - \frac{0}{10} = 0$$

$$\Rightarrow \frac{2V_1}{5} - \frac{V_2}{5} = 6$$

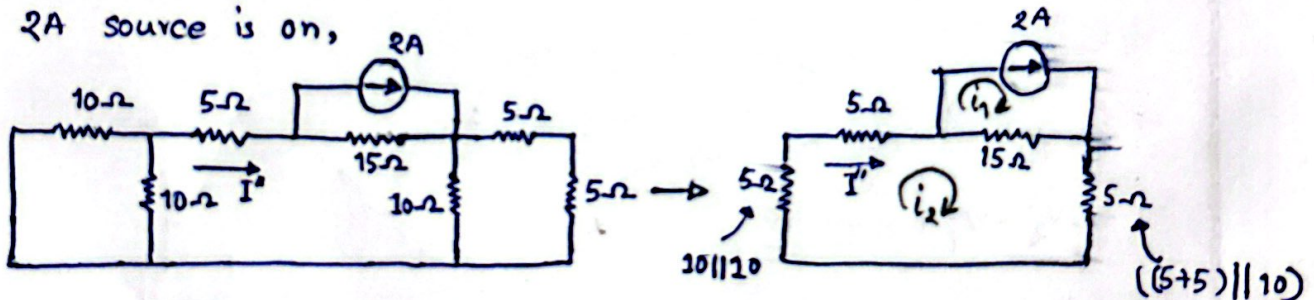
$$\Rightarrow 2V_1 - V_2 = 30 \quad \text{--- (1)}$$

Solving (1) and (2),

$$V_1 = 25V \quad V_2 = 20V$$

$$\text{Now, } I' = \frac{V_1 - V_2}{5} = \frac{25 - 20}{5} = 1A$$

Only 2A source is on,



$$\text{Loop 1, } i_1 = 2A$$

$$\text{KVL at Loop 2, } 5i_2 + 5i_2 + 15(i_2 - 2) + 5i_2 = 0$$

$$\Rightarrow 30i_2 = 30 \Rightarrow i_2 = 1A$$

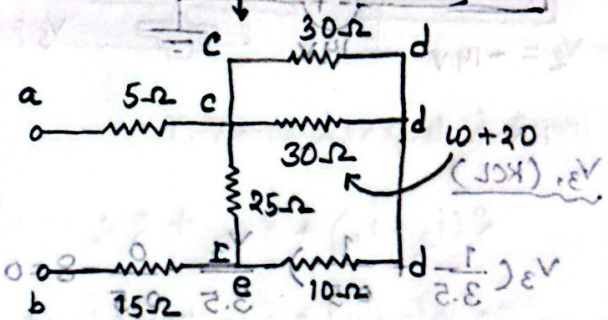
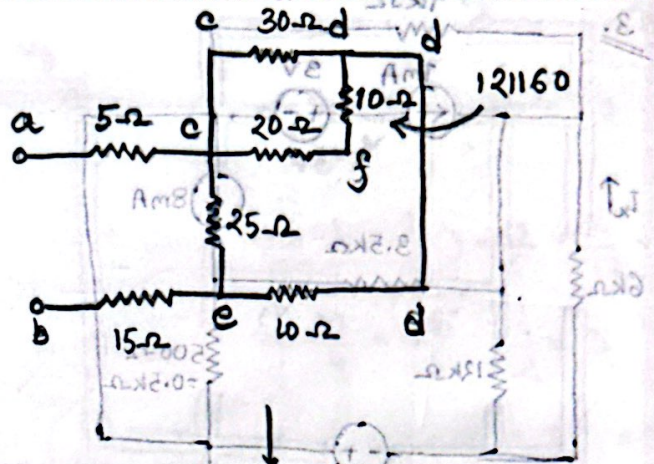
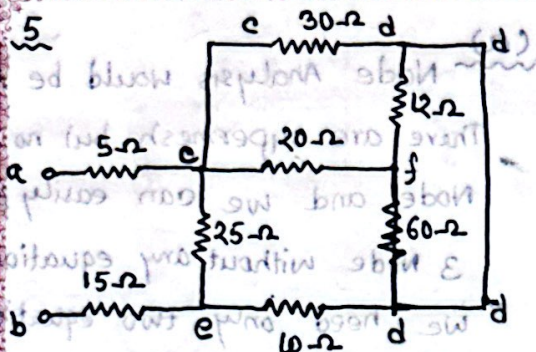
$$I'' = i_2 = 1A$$

Finally, $I = I' + I''$

$$= 1 + 1$$

$$= 2A$$

(Answer)



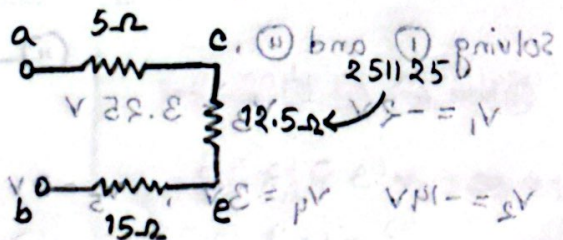
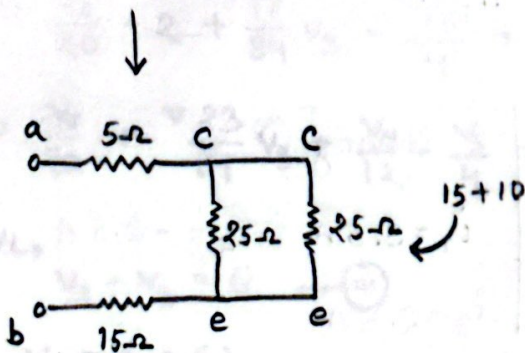
$$\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} + \frac{V}{R_4}$$

$$\frac{1}{R} = \frac{1}{30} + \frac{1}{10} + \frac{1}{25} + \frac{1}{15}$$

$$\frac{1}{R} = \frac{2}{60} + \frac{6}{60} + \frac{2.4}{60} + \frac{4}{60}$$

$$\frac{1}{R} = \frac{14.4}{60}$$

$$R = \frac{60}{14.4} = 4.166\Omega$$



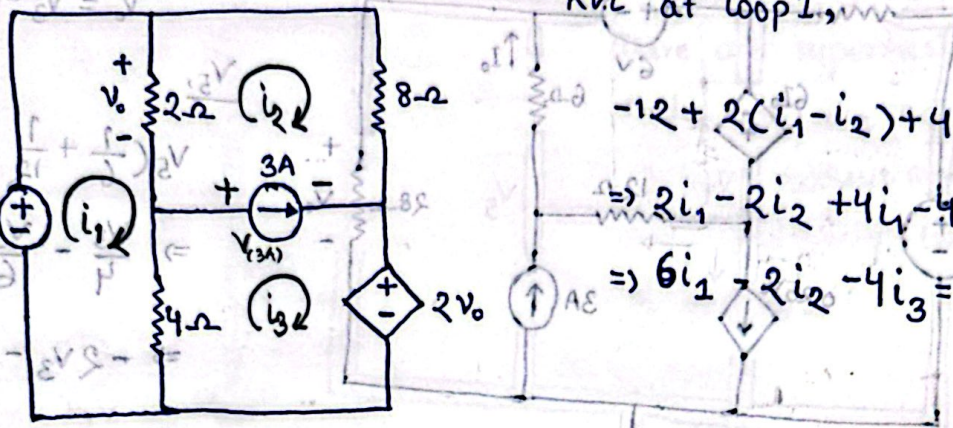
$$0 = \frac{V-2V}{15} + \frac{V-3V}{2.5} + I + \frac{V-2V}{1} + I$$

Answer: 32.5Ω

$$0 = \frac{5+11-}{15} + \frac{32.5-}{2.5} + \frac{5+0}{1} + I$$

$$5 + 12.5 + 15 \times I = 0$$

6



KVL at loop 1,

$$-12 + 2(i_1 - i_2) + 4(i_1 - i_3) = 0$$

$$\Rightarrow 2i_1 - 2i_2 + 4i_1 - 4i_3 = 12$$

$$\Rightarrow 6i_1 - 2i_2 - 4i_3 = 12 \quad \text{--- (i)}$$

loop 2 & loop 3 (supermesh)

$$2(i_2 - i_1) + 8i_2 + 2V_0 + 4(i_3 - i_1) = 0$$

$$\Rightarrow 2i_2 - 2i_1 + 8i_2 + 2(i_1 - i_2) + 4i_3 - 4i_1 = 0$$

$$\Rightarrow 2i_2 - 2i_1 + 8i_2 + 4i_1 - 4i_2 + 4i_3 - 4i_1 = 0$$

$$\Rightarrow -2i_1 + 6i_2 + 4i_3 = 0 \quad \text{--- (ii)}$$

SuperMesh Relation,

$$i_3 - i_2 = 3$$

$$\Rightarrow -i_2 + i_3 = 3 \quad \text{--- (iii)}$$

Solving (i), (ii) and (iii),

$$i_1 = 3.5 \text{ A} \quad i_2 = -0.5 \text{ A}$$

$$i_3 = 2.5 \text{ A}$$

Again, KVL at loop 2,

$$-V_{3A} + 2(i_2 - i_1) + 8i_2 = 0$$

$$\Rightarrow V_{3A} = 2(-0.5 - 3.5) + 8 \times (-0.5) = -2 - 4 = -6 \text{ V}$$

$$\therefore V_{3A} = -12 \text{ V}$$

(Answer)